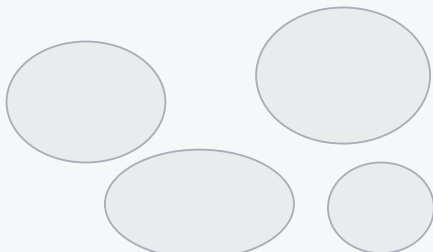


Why the pipeline diverged: a harder imaging regime

LIVER EM

vascular & cellular structures



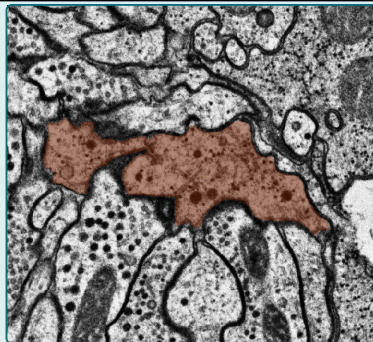
compact, convex, well separated

- compact, convex structures
- sparsely packed, easy to separate
- smooth across serial sections
- few instances per class
- dense organelles handled by nnU-Net
- run on an H100 GPU

VS

WORM EM

C. elegans neuropil



one thin neurite (salmon) among crowded neighbours

- thin neurites, a few px wide at scale-8
- densely packed, many touching
- branch / merge points break tracking
- ~300 neurons, thousands of chains
- single consumer GPU
- all-SAM2 (no second dense model yet)

Thinner, more crowded, more numerous, and more branched. Every addition in the pipeline diff (chain decomposition, negative prompts, tiered crops, QC, triage, cross-worm evaluation) is a response to this regime shift.